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tfl_mgh_multiecho																	

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\rslh_ep3d_vaso

TA: 14 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---

Contrast - Common

TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Multi-echo spacing	88.00 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	3130.32 ms
TI 2	9370.32 ms
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vso rslh6.0
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.04 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1080 us
RF time x BW	18
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
Invert PE	Off
Min. TE w/ PF	On
Expert ramping	Off
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Phase Correction	per Series
EPI ramp factor	1.10
EPI ramp factor 2	1.10
Ramp sampling	1.00 1:max
Slab Scale	0 %
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off
GRAPPA Regularization	5000 /10^6
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Read polarity	Regular RO
VASO Pre-T11 delay	0.00 ms
VASO Pre-T12 delay	0 ms
VASO Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\ZPL_RG_EPSI_SE_v1b

TA: 1:23 min Coil Selection: Auto Voxel Size: 24.0×4.0×12.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	100 mm
TR	800.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	800.0 ms
TE	1.60 ms
Ernst angle	56 deg
WaterSupp Angle	90 deg
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um
Flip Angle	90 deg
Preparation Scans	2
Averages	1
Water Suppression	
Water Suppr. BW	80 Hz

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Scan Res. A >> P	10
Scan Res. R >> L	60
Scan Res. F >> H	10
Interpol. Res. A >> P	128

Resolution - Common

Interpol. Res. R >> L	128
Interpol. Res. F >> H	64
Hamming	Off
Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	120 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	100 mm
Fully Excited Vol	Off
Saturation Region	1
Thickness	100.00 mm
Position	L0.0 P0.0 H104.7 mm
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm
Saturation Region	2
Thickness	100.00 mm
Position	L0.0 P0.0 H104.7 mm
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Transversal
! Rotation	0.00 deg
! A >> P	240 mm
! R >> L	240 mm
! F >> H	100 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsilon_se_v1b
Preparation Scans	2
Delta Frequency	-2.40 ppm
Dimension	3D
Save Uncombined	Off
Acquisition Delay	2.30 ms
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp Pattern	No Customized Sampling Pattern
Mode. Bipolar	1 #
Mode. YreadNav	0 #
Mode: RFspoil	1 #
Ramp. Samp. EPSI	0 #
Ramping time (RO)	110 us
Ramping time (PE)	120 us
Grad duration (PE)	600 us

Sequence - Special

Blip ramp time (PE)	60 us
Blip ramp time (SL)	60 us
Grad. EchoSpace(RO)	1160 us
Grad. flat time(RO)	360 us
Readout ADC (RO)	360 us
ADC dwell time	6 us
Acq. advance time	0 us
Refocussing pulse voi	160 %
Crusher duration (Cr)	1200 us
Num: Echo-Shift	0 #
Number of TEs	2 #
TE values[1]	50 ms
TE values[2]	190 ms
TE values[3]	420 ms
TE values[4]	620 ms
TE values[5]	820 ms
TE values[6]	1020 ms
TE values[7]	1220 ms
TE values[8]	1420 ms
Echo-Pairs. EPSI[1]	54 #
Echo-Pairs. EPSI[2]	256 #
Echo-Pairs. EPSI[3]	40 #
Echo-Pairs. EPSI[4]	40 #
Echo-Pairs. EPSI[5]	40 #
Echo-Pairs. EPSI[6]	40 #
Echo-Pairs. EPSI[7]	40 #
Echo-Pairs. EPSI[8]	40 #
Echos of FID	14 #
Echos before TEs[1]	16 #
Echos before TEs[2]	54 #
Echos before TEs[3]	0 #
Echos before TEs[4]	0 #
Echos before TEs[5]	0 #
Echos before TEs[6]	0 #
Echos before TEs[7]	0 #
Echos before TEs[8]	0 #
Crusher amplitude1(Cr)[1]	16 us
Crusher amplitude1(Cr)[2]	8 us
Crusher amplitude1(Cr)[3]	16 us
Crusher amplitude1(Cr)[4]	16 us
Crusher amplitude1(Cr)[5]	16 us
Crusher amplitude1(Cr)[6]	16 us
Crusher amplitude1(Cr)[7]	16 us
Crusher amplitude1(Cr)[8]	16 us
Fnav. read points	100 #
Fnav. dwell time	10 us

\\Research\Investigators\Frederick\CORPUS_CATALOGED\cmrr_mbep2d_bold

TA: 13:16 min Coil Selection: Manual Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
TE	37.00 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HC1-7

Contrast - Common

TR	800.0 ms
TE	37.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	983
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	800.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	800.0 ms
Log Signals	Off
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	983
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Flow Compensation	None
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	Off
EPI Factor	104

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	6600 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\cmrr_mbep2d_diff

TA: 6:05 min Coil Selection: Auto Voxel Size: 2.1×2.1×1.5 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Resolution - Common

Slice Thickness	1.5 mm
Base Resolution	98
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Routine

Slice Group	1
Slices	92
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	3230.0 ms
TE	89.20 ms
Multi-band accel. factor	4
AutoAlign	Head > Brain

Geometry - Common

Slice Group	1
Slices	92
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	1.5 mm
TR	3230.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	4

Contrast - Common

TR	3230.0 ms
TE	89.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	78 deg
Refocus flip angle	160 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H2.0
L	0.0 mm
P	0.0 mm
H	2.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	2 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	138 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3230.0 ms
Multi-band accel. factor	4

Physio - PACE

Resp. Control	Off
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Physio - PACE

Multi-band accel. factor	4
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Diff

Diffusion Mode	Free
Diff. Directions	107
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	3000 s/mm ²
Averages 1	1
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1700 Hz/Px
Echo Spacing	0.69 ms
Free Echo Spacing	Off
EPI Factor	98

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	3840 us
Refocus pulse duration	7680 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	On
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off

Sequence - Special

Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	DICOM

Sequence - Assistant

SAR Assistant	Off
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\\Research\Investigators\Frederick\CORPUS_CATALOGED\cmrr_mbep2d_se

TA: 32 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain

Contrast - Common

TR	8036.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	3
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	104
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	8036.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	T > C-16.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	T > C
T > C	-16.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	T > C-16.0
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8036.0 ms
Log Signals	Off
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	3
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Performance
Bandwidth	2290 Hz/Px
Echo Spacing	0.58 ms
Free Echo Spacing	On
EPI Factor	104

Sequence - Part 2

Introduction	Off
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Sequence - Special

Min. prep scans	0
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_csi_laser

TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms

Sequence - Special

VAPOR delay 1	150 ms
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_csi_mslaser

TA: 25:57 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us

Sequence - Special

OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_csi_press

TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40

Sequence - Special

OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_csi_slaser

TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1

Sequence - Special

OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_svs_laser

TA: 2:00 min Coil Selection: Auto Vol: 25×25×25 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	25 mm
Vol R >> L	25 mm
Vol F >> H	25 mm
TR	3000.0 ms
TE	74.52 ms
Averages	32

Contrast - Common

TR	3000.0 ms
TE	74.52 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	32
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	25 mm
Vol R >> L	25 mm
Vol F >> H	25 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	25 mm
R >> L	25 mm
F >> H	25 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser
---------------	-------

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Resolve averages	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_svs_mpress

TA: 6:46 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	2000.0 ms
TE	68.00 ms
Averages	96

Contrast - Common

TR	2000.0 ms
TE	68.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	96
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	mpres
---------------	-------

Sequence - Common

Preparation Scans	4
Delta Frequency	-2.80 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	On
MEGA water suppr.	Off
Inversion pulse	Off

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_svs_mslaser

TA: 6:51 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr
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Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_svs_press

TA: 3:01 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	2000.0 ms
TE	30.00 ms
Averages	80

Contrast - Common

TR	2000.0 ms
TE	30.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	80
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	80.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	1024
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2000.0 ms

Sequence - Common

Sequence Name	press
---------------	-------

Sequence - Common

Preparation Scans	4
Delta Frequency	-2.30 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	1200 Hz
Acquisition Duration	853 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\\Investigators\\Frederick\\CORPUS_CATALOGED\\eja_svs_slaser

TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr
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Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Shift RO frequency	Off
Debug loop type	None

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off

\\Research\Investigators\Frederick\CORPUS_CATALOGED\leja_svs_steam

TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam
---------------	-------

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators\Frederick\CORPUS_CATALOGED\tfl_mgh_multiecho

TA: 6:30 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	60 %
Slice Resolution	100 %
Interpolation	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HC1-7

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2530.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Contrast - Common

TR	2530.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Geometry - AutoAlign

Slab Group	1
Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal

Geometry - AutoAlign

Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R2.2 P1.7 F31.8
R	2.2 mm
P	1.7 mm
F	31.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Table Position	32 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	On
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R2.2 P1.7 F31.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.249408 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	2530.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FOV Read	256 mm
FOV Phase	100.0 %
Phase Resolution	60 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	tfl_me
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px

Sequence - Part 1

Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	176

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Readout trajectory	Bipolar
Gradient spoiling	Siemens
Gradient moment factor	1.00
Averaging	RMS

Sequence - Assistant

SAR Assistant	Off
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